The empirical recurrence for OEIS A233101, proved by carrying a row major clause across columns

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Abstract

OEIS A233101 counts $4 \times n$ arrays over $\{0, \dots, 3\}$ with a forbidden-pair condition and a clause on the order in which values first appear, read in ROW MAJOR order, and it carries an empirical recurrence of order 11 contributed by R. H. Hardin. It is true. The obstacle is that the array grows in n, so the walk runs across COLUMNS and does not visit the cells in row major order; the running set of "values seen so far" that settles the clause for a column-fixed entry is simply the wrong set here. What the walk can carry instead is, for each value the clause names, the least ROW in which it has yet been seen, together with the order in which those least rows were attained. That is enough, and it is bounded, so the count is a walk count and the recurrence is settled by a finite exact computation.

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1 The conjecture

OEIS A233101 is "Number of 4Xn 0..3 arrays with no element x(i,j) adjacent to value 3-x(i,j) horizontally, diagonally or antidiagonally, top left element zero, and 1 appearing before 2 in row major order.". It has offset 1 and begins

$$36, 515, 13721, 342626, 8714705, 221113913, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 25*a(n-1) +52*a(n-2) -1084*a(n-3) +16*a(n-4) +9754*a(n-5) -16165*a(n-6) +5229*a(n-7) +5498*a(n-8) -3740*a(n-9) +320*a(n-10) +96*a(n-11) for n>12

As of the "Last modified" line on the live entry (Jul 23 2025 07:52:46, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 4 rows and n columns. Two cells are adjacent when they differ by one of the offsets

$$(0, 1), (1, -1), (1, 1),$$

and the entry forbids any adjacent pair u, v with $u + v = 3$. It fixes the top left cell to zero. Finally it asks that, reading the cells in ROW MAJOR order — along the first row, then along the second, and so on — the first occurrences of the values 1, 2 come in a prescribed order: 1 before 2. A precedence involving a value that never occurs is satisfied vacuously; one whose later value occurs while its earlier value does not is broken.

3 Why the obvious state does not work

Since n grows, the walk must run across the columns: a state is one column, and the forbidden-pair condition is settled between two consecutive columns, because every offset above moves the column index by at most one.

The clause is the difficulty. For an entry whose WIDTH is fixed the walk runs down the rows and therefore visits the cells in row major order, so it is enough to carry the set of named values already written and to refuse a column that introduces a value out of turn. Here the walk visits the cells in COLUMN major order, and that set is the wrong one: a value first written low down in an early column may be written again higher up in a later column, and it is the later, higher cell that comes first in row major order.

4 What the walk can carry

Lemma 1. *For a value v occurring in the array, let r_v be the least row in which v occurs and c_v the least column with $x_{r_v,c_v} = v$. Then the first occurrence of v in row major order is the cell (r_v, c_v), and u precedes v in row major order exactly when $(r_u, c_u) < (r_v, c_v)$ lexicographically.*

Proof. Row major order is the lexicographic order on (row, column), and the least element of the set of cells carrying v under a lexicographic order is obtained by minimising the first coordinate and then the second. □

Lemma 2. *Two distinct values with the same least row attain it in different columns.*

Proof. If $r_u = r_v = r$ and $c_u = c_v = c$ then the cell (r, c) carries both u and v. □

So the walk carries, for each named value, the least row in which it has been seen so far — or the fact that it has not been seen — and, for values sharing a least row, which of them attained it first. By Lemma 2 those attainments happen in different columns, and the walk meets the columns in order, so "which attained it first" is decided as the walk goes. Nothing else about the past matters: a later column can only LOWER a value's least row, and Lemma 1 says the comparison depends on nothing else.

One thing must not be done, and is not done here. The clause cannot be enforced along the way, because a least row can still drop and reorder the first occurrences retroactively. It is a property of the finished array, so it is tested in the ACCEPTING states and nowhere else.

The state is therefore a column together with an ordered partition of the named values by least row, and there are $S = 163$ of them after states with identical futures are merged. Hence

$$a(n) \;=\; \iota^{\top} M^{n-1} \tau,$$

and a is C-finite.

5 The proof

Lemma 3. *Let $q(t) = t^{11} - \sum_i c_i t^{11-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^j q(M) \tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M) \tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+11} - \sum_i c_i M^{j+11-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 4.

$$a(n) = 25\,a(n-1)+52\,a(n-2)-1084\,a(n-3)+16\,a(n-4)+9754\,a(n-5)-16165\,a(n-6)+5229\,a(n-7)+5498\,a(n$$

for every $n > 12$.

Proof. The vector $w = q(M)\tau$ was formed by 11 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots$, again exactly; those integers vanish from the index corresponding to $n = 12 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. The bound is exact: at $n = 12$ the recurrence fails on the entry's own published terms, so no larger range holds. □

6 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic, with no shift fitted.

Second — and this is the check the whole paper turns on — the arrays were enumerated directly for the smallest shapes the entry counts, in the orientation the name writes, with the cells listed in true row major order and each precedence tested on that list. No transposing and no walk. The counts agree. Reading the clause in the walk's own order instead gives different numbers for several entries of this family, so this check is what separates the two readings.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 3 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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